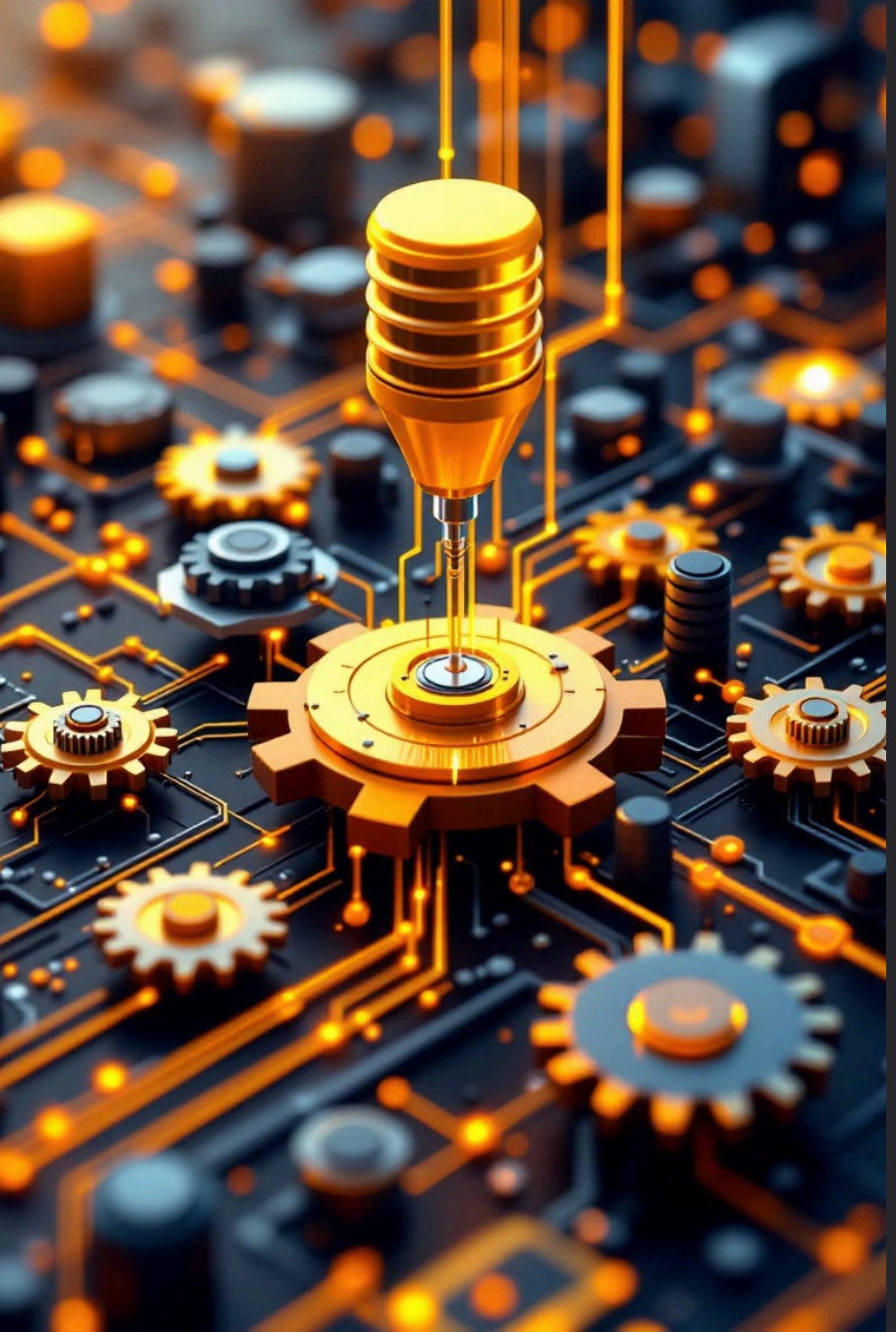

Configuration Management: introduzione ad Ansible

Hands On Workshop

by Evanghelos Papaioannu

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Welcome and pleased to meet you!

- Let's make it easy: Evan
- Worked in some big companies (VF, IBM, YNAP)
- Mostly in Operations and mission critical environments
- Linux System Administrator > DevOps > SRE
  - System Design
  - Observability
  - AI

- Love outdoor life with family
- Loved to hike
- Loved to bike
- Loved to travel

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What is Configuration Management?

Configuration management tools automate and standardize system setups across your infrastructure.

Consistency

Identical environments across development, testing, and production reduce "works on my machine" issues.

Version Control

Track and roll back changes with complete infrastructure history.

Scalability

Deploy coherent configurations to thousands of servers with minimal effort.

Cattle vs Pet.

Compliance

Enforce security policies and meet regulatory requirements consistently.

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Configuration Management in the IT Lifecycle

Configuration Management bridges development and operations, enabling infrastructure consistency throughout the entire IT lifecycle.

Infrastructure as Code

Define infrastructure through version-controlled code rather than manual processes.

Iteration & Evolution

Refine infrastructure based on operational insights and new requirements.

Configuration Management

Maintain system state and enforce desired configurations across environments.

Continuous Deployment

Automate testing and deployment of infrastructure changes.

Monitoring & Feedback

Track performance and compliance to inform optimization efforts.

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Cognitive Load in IT and Automation Benefits

The Cognitive Burden

Managing complex IT infrastructures requires constant mental effort and attention.

- Manual configurations demand deep technical knowledge
- Tracking changes across environments causes mental fatigue
- Context-switching between tools fragments attention

Automation Relief

Configuration management tools like Ansible offload repetitive mental tasks.

- Codified processes require less working memory
- Templated solutions reduce decision fatigue
- Self-documenting infrastructure enhances knowledge sharing

Operational Benefits

The cognitive savings translate to tangible IT lifecycle improvements.

- More time for innovation and strategic thinking
- Reduced errors in critical environments
- Faster response to changing requirements

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Configuration Management Alternatives

Chef

Ruby-based tool with a focus on continuous automation.

- Uses "recipes" and "cookbooks"
- Pull architecture
- Steep learning curve

Puppet

Model-driven approach with declarative language.

- Custom DSL syntax
- Strong reporting tools
- Mature enterprise support

SaltStack

Python-based with event-driven automation.

- High-speed communication
- YAML for configurations
- Strong remote execution

Ansible

Agentless with simple YAML syntax.

- No agents required
- Low learning curve
- SSH-based communication

While **IaC tools like Terraform and Pulumi** focus on provisioning infrastructure (creating servers, networks), **configuration management** tools maintain the state of existing resources. IaC builds the foundation; CM tools shape what runs on it.

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Key Ansible Features

Agentless

Uses SSH (Linux) or WinRM (Windows) to execute tasks remotely without requiring agents on managed nodes.

Idempotency

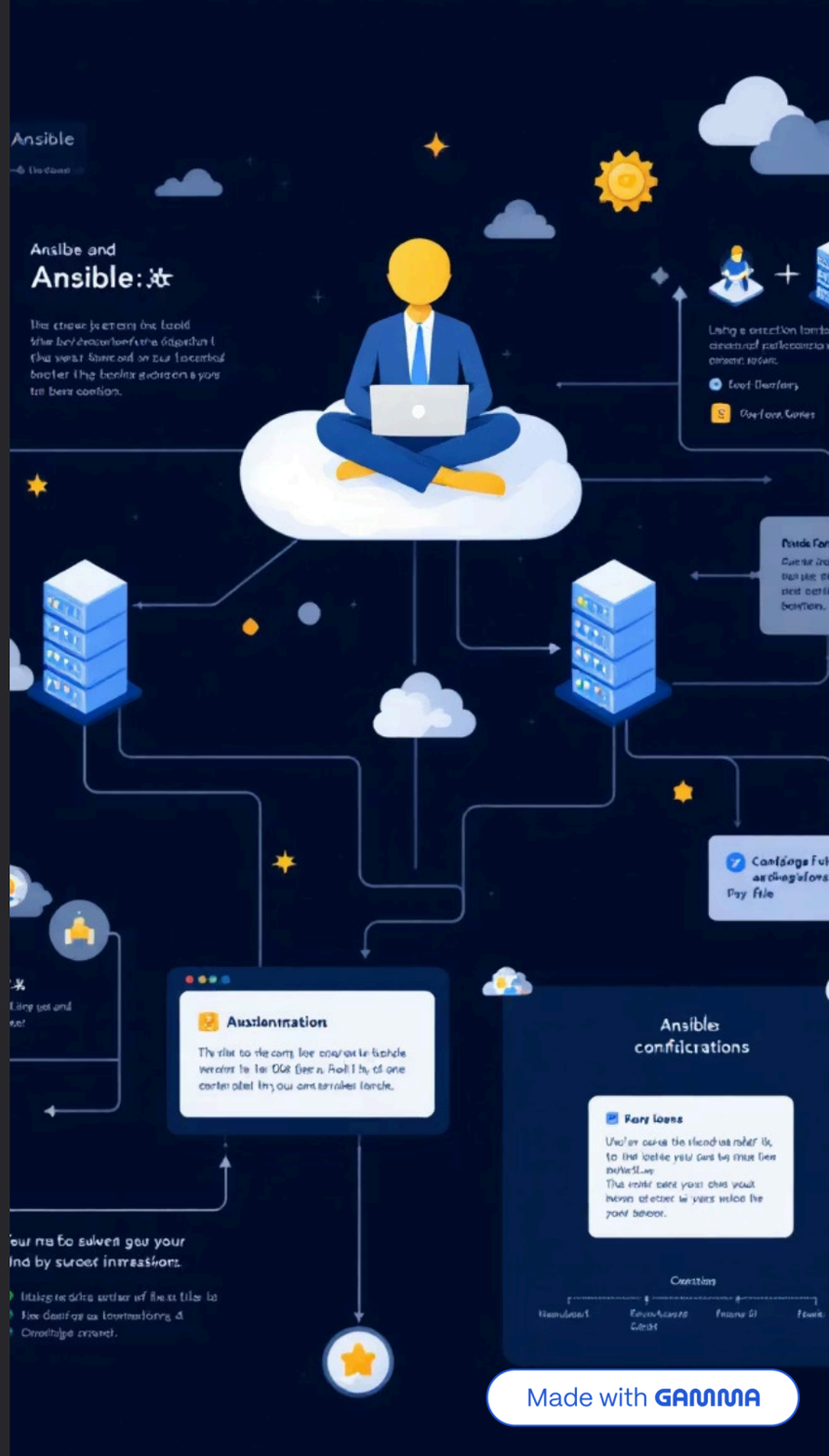
Running a playbook multiple times won't cause unintended side effects; changes are only applied if needed.

Inventory Management

Manages dynamic and static inventories from various sources like cloud providers or databases.

Event-driven Automation

Responds to events and integrates with tools like AWX/Ansible Tower for automation workflows.



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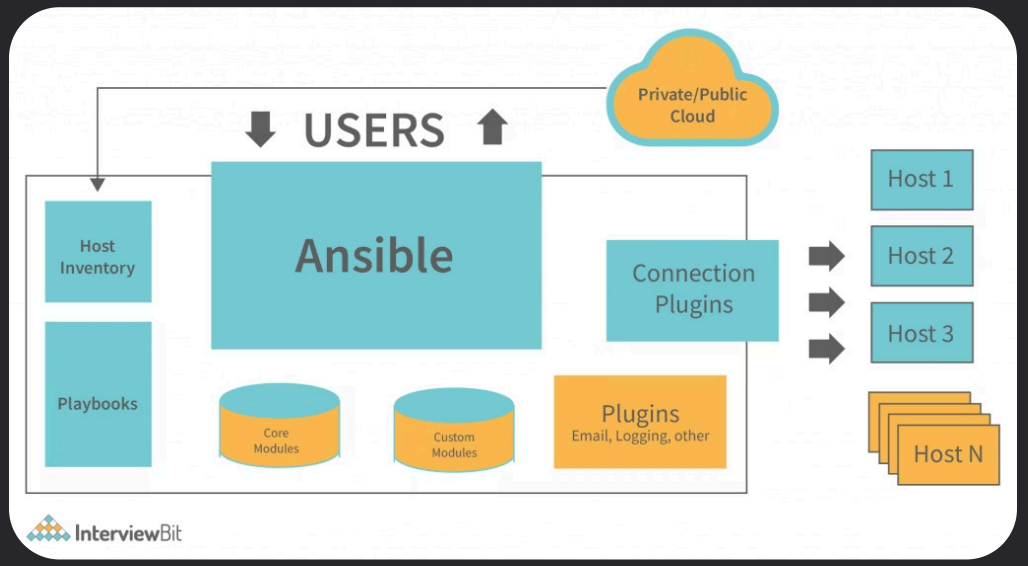
Ansible Architecture Overview

- Inventory
- Tasks
- Playbook
- (Roles)
- (Modules)



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Hands on!

git clone

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Understanding Ansible Inventory

Inventory Formats

- INI format
- YAML format
- JSON format

Inventories help manage servers at scale - from tens to thousands of servers.

Grouping Servers

Servers can be organized in groups like:

- [web] for web servers
- [db] for database servers
- [demo] for testing environments

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Ansible Facts

System Properties

Automatically gathered system information including OS, network interfaces, CPU, and memory details.

Gathering Facts

Use the setup module:

ansible -i hosts hostname -m setup

Filtering Facts

Filter specific information:

ansible -i hosts hostname -m setup -a "filter=ansible_distribution*"

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Ansible Galaxy

- **Namespaces**
  - ansible
  - community
  - amazon
  - azure
- **Collections**
  - Modules (idempotent units of work)
  - Plugins (callback, connection, lookup, filter)
  - Roles (reusable task/variable/file structures)
  - Playbooks (example/playbook YAML files bundled for demos or integration tests)

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Playbook

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Automation: Craft, Not Replacement

The Automation Misconception

Many fear automation means job elimination and reduced quality.

This view misses how automation transforms rather than replaces human work.

- Changes skill requirements
- Shifts focus to higher-level thinking
- Eliminates repetitive tasks

The IT Artisan's Approach

Modern IT engineers are digital craftspeople creating automation solutions.

They blend technical expertise with creative problem-solving.

- Designs resilient systems
- Creates elegant automation workflows
- Builds with precision and care

Elevated Capabilities

Automation brings new competencies to the forefront.

Engineers who embrace these skills deliver superior results.

- Systems thinking
- Architecture design
- Continuous improvement

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Resources

ANSIBLE

docs.ansible.com

Ansible Documentation

SEMAPHORE UI

Modern UI for Ansible

Semaphore UI for Ansible,...

Modern UI and powerful AP...

If your project has outgrown manual terminal deployments,...

Cattle vs Pet

cloudscaling

Cloudscaling

The History of Pets vs Cattl...

I have been meaning to write this post for a long time, but one...

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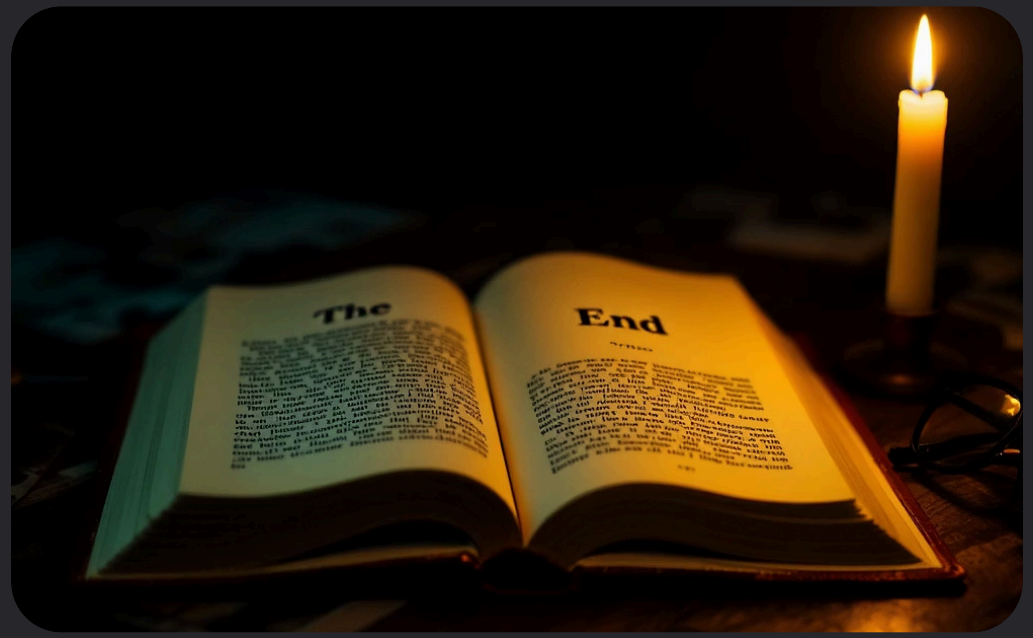
The End



Closing thoughts

Q&A

Thanks for your time!

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